

NEIL SHAH

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EDUCATION

- **PhD:** Computer Science, Carnegie Mellon University, August 2013 - October 2017
Advisor: Prof. Christos Faloutsos. Degree received: December 20, 2017.
- **MS:** Computer Science, Carnegie Mellon University, August 2013 - May 2017
- **BS:** Computer Science (Minor in Mathematics), North Carolina State University, August 2010 - May 2013
GPA: **4.0** (class rank #1), *Summa Cum Laude with Honors*.

POSITIONS

- **Snap Inc.**, Director of Research, Senior Principal Scientist.
Lead a horizontal team of researchers and engineers on academic, applied research, and engineering initiatives in user modeling and personalization, with applications to growth, content, ads, and safety.
December 2017 - present
- **Carnegie Mellon University**, Graduate Researcher.
Computer Science Department. Worked on algorithms and applications for anomaly detection in large social graphs.
August 2013 - October 2017
- **Twitch**, Visiting Researcher.
Worked on anti-abuse technologies as a member of the Science team.
January 2016 - May 2016
- **Microsoft Research**, Research Intern.
Improved metrics and methods for measuring research impact for Microsoft Academic Search.
June 2015 - August 2015
- **Lawrence Livermore National Laboratory**, Research Intern.
Developed algorithms to automatically identify patterns and anomalies in time-evolving graphs.
June 2014 - August 2014
- **IBM Silicon Valley Lab**, Software Intern.
Worked in the IBM BigInsights group on indexing and analytics of system log data.
May 2012 - January 2013
- **North Carolina State University**, Undergraduate Researcher.
Department of Computer Science. Worked on compressing and indexing large scientific datasets.
June 2009 - April 2013

AWARDS & DISTINCTIONS

- ACM CIKM Best Paper Award, 2025
- North Carolina State University Department of Computer Science Rising Star Award, 2023
- ACM WSDM Outstanding Service Award, 2022
- ACM SIGCHI Best Research Paper Honorable Mention Award, 2019
- Symantec Graduate Research Fellowship Finalist, 2017
- ACM SIGKDD Best Research Paper Award, 2016
- National Science Foundation Graduate Research Fellowship, 2013
- North Carolina State University College of Engineering Senior Award for Scholarly Achievement, 2013

- North Carolina State University Department of Computer Science Senior Faculty Scholar, 2012
- National Science Foundation Research Experience for Undergraduates Grant, 2011
- North Carolina State University College of Engineering Dean’s Research Assistantship, 2011
- North Carolina State University Caldwell Fellowship, 2011
- Coca-Cola Scholarship, 2010
- Zinch Scholarship, 2010
- National Merit Scholarship, 2010
- CompTIA Information Technology Merit Award, 2010
- 2nd place, National Siemens Competition in Math, Science and Technology, 2009
- 1st place, Regional Siemens Competition in Math, Science and Technology, 2009

PUBLICATIONS

Conference Publications

95. H. Zeng, L. Collins, B. Kumar, N. Shah, H. Zamani. *CoSearch: Joint Training of Reasoning and Document Ranking via Reinforcement Learning for Agentic Search*. **COLM 2026**.
94. L. Collins, B. Kumar, C. Ju, T. Zhao, D. Loveland, L. Neves, N. Shah. *Understanding ID-Text Complementarity in Sequential Recommendation*. **RecSys 2026**.
93. J. Liu, H. Wang, J. Tang, L. Collins, T. Zhao, N. Shah, C. Ju. *Beyond Fixed Depths and Widths: Optimizing Textual Decoding Tries in LLM-based Generative Recommendation*. **RecSys 2026**.
92. J. Liu, L. Collins, J. Tang, T. Zhao, N. Shah, C. Ju. *Understanding Generative Recommendation with Semantic IDs from a Model Scaling View*. **KDD 2026**.
91. X. Ding, X. Huang, C. Ju, L. Collins, Y. Liu, L. Akoglu, N. Shah, T. Zhao. *Hierarchical Token Prepending: Enhancing Information Flow in Decoder-based LLM Embeddings*. **ACL 2026**.
90. X. Huang, X. Ding, M. Ju, Y. Liu, N. Shah, T. Zhao. *Threshold Differential Attention for Sink-Free, Ultra-Sparse, and Non-Dispersive Language Modeling*. **ACL 2026**.
89. J. Zhu, M. Ju, Y. Liu, S. Vij, D. Koutra, N. Shah, T. Zhao. *Beyond Unimodal Perspectives: Generative Retrieval with Multimodal Semantics*. **SIGIR 2026**.
88. C. Ju, T. Zhao, L. Neves, L. Collins, B. Kumar, et al. *Semantic IDs for Recommender Systems at Snapchat: Use Cases, Technical Challenges, and Design Choices*. **SIGIR 2026**.
87. Q. Truong, Y. Song, D. Loveland, C. Ju, T. Zhao, N. Shah, J. Tang. *Plain Transformers are Surprisingly Powerful Link Predictors*. **ICML 2026**.
86. G. Lee, B. Kumar, C. Ju, T. Zhao, K. Shin, N. Shah, L. Collins. *Sequential Data Augmentation for Generative Recommendation*. **WSDM 2026**.
85. Z. Zheng, Y. Zhu, H. Liu, C. Ju, T. Zhao, N. Shah, J. Li. *Pretrained Language Model based Cold-Start Recommendation with Meta-Item Embeddings*. **CIKM 2025**.
84. C. Ju, L. Collins, L. Neves, B. Kumar, L. Wang, T. Zhao, N. Shah. *Generative Recommendation with Semantic IDs: A Practitioner’s Handbook*. **CIKM 2025. Best Paper Award**.
83. Q. Truong, Z. Chen, C. Ju, T. Zhao, N. Shah, J. Tang. *A Pre-Training Framework for Relational Data with Information Theoretic Principles*. **NeurIPS 2025**.
82. D. Loveland, M. Ju, T. Zhao, N. Shah, D. Koutra. *On the Role of Weight Decay in Collaborative Filtering: A Popularity Perspective*. **KDD 2025**.
81. T. Zhao, Y. Liu, M. Kolodner, K. Montemayor, E. Ghazizadeh, et al. *GiGL: Large-Scale Graph Neural Networks at Snapchat*. **KDD 2025**.
80. M. Ju, L. Neves, B. Kumar, L. Collins, T. Zhao, et al. *Revisiting Self-Attention for Cross-Domain Sequential Recommendation*. **KDD 2025**.
79. M. Ju, L. Neves, B. Kumar, L. Collins, T. Zhao, et al. *Learning Universal User Representations Leveraging Cross-domain User Intent at Snapchat*. **SIGIR 2025**.
78. R. Xue, T. Zhao, N. Shah, X. Liu. *Haste Makes Waste: A Simple Approach for Scaling Graph Neural Networks*. **ICML 2025**.

77. N. Bui, M. Yang, R. Chen, L. Neves, C. Ju, R. Ying, N. Shah, T. Zhao. *Learning Along the Arrow of Time: Hyperbolic Geometry for Backward-Compatible Representation Learning*. **ICML 2025**.
76. D. Loveland, X. Wu, T. Zhao, D. Koutra, N. Shah, M. Ju. *Understanding and Scaling Collaborative Filtering Optimization from the Perspective of Matrix Rank*. **WWW 2025**.
75. X. Wu, D. Loveland, R. Chen, Y. Liu, X. Chen, et al. *GraphHash: Graph Clustering Enables Parameter Efficiency in Recommender Systems*. **WWW 2025**.
74. J. Zhu, Y. Zhou, S. Qian, Z. He, T. Zhao, N. Shah, D. Koutra. *Mosaic of Modalities: A Comprehensive Benchmark for Multimodal Graph Learning*. **CVPR 2025**.
73. J. Liu, H. Mao, Z. Chen, T. Zhao, N. Shah, J. Tang. *Towards Neural Scaling Laws on Graphs*. **LoG 2024**.
72. M. Ju, W. Shiao, Z. Guo, Y. Ye, Y. Liu, N. Shah, T. Zhao. *Test-time Aggregation For Collaborative Filtering*. **NeurIPS 2024**.
71. P. Kung, Z. Fan, T. Zhao, Y. Liu, Z. Lai, J. Shi, Y. Wu, J. Yu, N. Shah, G. Venkataraman. *Improving Embedding-Based Retrieval in Friend Recommendation with ANN Query Expansion*. **SIGIR 2024**.
70. R. Chen, T. Zhao, A. K. Jaiswal, N. Shah, Z. Wang. *LLaGA: Large Language and Graph Assistant*. **ICML 2024**.
69. H. Mao, Z. Chen, W. Tang, J. Zhao, Y. Ma, T. Zhao, N. Shah, M. Galkin, J. Tang. *Graph Foundation Models*. **ICML 2024**.
68. A. Calabrese, L. Neves, N. Shah, M. Bos, B. Ross, M. Lapata, F. Barbieri. *Explainability and Hate Speech: Structured Explanations Make Social Media Moderators Faster*. **ACL 2024**.
67. H. Mao, J. Li, H. Shomer, B. Li, W. Fan, Y. Ma, T. Zhao, N. Shah, J. Tang. *Revisiting Link Prediction: A Data Perspective*. **ICLR 2024**.
66. Y. Wang, T. Zhao, Y. Zhao, Y. Liu, X. Cheng, N. Shah, T. Derr. *A Topological Perspective on Demystifying GNN-based Link Prediction Performance*. **ICLR 2024**.
65. T. Zhao, N. Shah, E. Ghazizadeh. *Learning from Graphs without Explicit Graph Machine Learning Models*. **ICLR 2024**.
64. J. Li, H. Shomer, H. Mao, S. Zeng, Y. Ma, N. Shah, J. Tang, D. Yin. *Evaluating Graph Neural Networks for Link Prediction: Current Pitfalls and New Benchmarking*. **NeurIPS 2023**.
63. H. Mao, Z. Chen, W. Jin, H. Han, Y. Ma, T. Zhao, N. Shah, J. Tang. *Demystifying Structural Disparity in Graph Neural Networks: Can One Size Fit All?*. **NeurIPS 2023**.
62. M. Ju, T. Zhao, W. Yu, N. Shah, F. Ye. *GraphPatcher: Mitigating Degree Bias for Graph Neural Networks via Test-time Node Patching*. **NeurIPS 2023**.
61. W. Shiao, U. S. Saini, Y. Liu, T. Zhao, N. Shah, E. Papalexakis. *CARL-G: Clustering-Accelerated Representation Learning on Graphs*. **KDD 2023**.
60. S. Bhatia, M. Wadhwa, K. Kawaguchi, N. Shah, P. S. Yu, B. Hooi. *Sketch-based Anomaly Detection in Streaming Graphs*. **KDD 2023**.
59. Z. Guo, W. Shiao, S. Zhang, Y. Liu, N. Chawla, N. Shah, T. Zhao. *Linkless Link Prediction via Relational Distillation*. **ICML 2023**.
58. J. Shi, V. Chaurasiya, Y. Liu, S. Vij, Y. Wu, S. Kanduri, N. Shah, P. Yu, N. Srivastava, L. Shi, G. Venkataraman, J. Yu. *Embedding-based Retrieval in Friend Recommendation*. **SIGIR 2023**.
57. J. Li, H. Shomer, J. Ding, Y. Wang, Y. Ma, N. Shah, J. Tang, D. Yin. *Are Message Passing Neural Networks Really Helpful for Knowledge Graph Completion?*. **ACL 2023**.
56. X. Han, T. Zhao, Y. Liu, X. Hu, N. Shah. *MLPInit: Embarrassingly Simple GNN Training Acceleration with MLP Initialization*. **ICLR 2023**.
55. M. Ju, T. Zhao, Q. Wen, W. Yu, N. Shah, Y. Ye, C. Zhang. *Multi-task Self-supervised Graph Neural Networks Enable Stronger Task Generalization*. **ICLR 2023**.
54. W. Shiao, Z. Guo, T. Zhao, V. Papalexakis, Y. Liu, N. Shah. *Link Prediction with Non-Contrastive Learning*. **ICLR 2023**.
53. W. Jin, T. Zhao, J. Ding, Y. Liu, J. Tang, N. Shah. *Empowering Graph Representation Learning with Test-Time Graph Transformation*. **ICLR 2023**.
52. Y. Wang, B. Hooi, Y. Liu, T. Zhao, Z. Guo, N. Shah. *Flashlight: Scalable Link Prediction with Effective Decoders*. **LoG 2023**.
51. Y. Wang, B. Hooi, Y. Liu, N. Shah. *Graph Explicit Neural Networks: Explicitly Encoding Graphs for Efficient and Accurate Inference*. **WSDM 2023**.

50. R. Baten, Y. Liu, H. Peters, F. Barbieri, N. Shah, L. Neves, M. Bos. *Predicting Future Location Categories of Users in a Large Social Platform*. **ICWSM 2023**.
49. S. Zhang, Y. Liu, N. Shah, Y. Sun. *Explaining Graph Neural Networks with Structure-Aware Cooperative Games*. **NeurIPS 2022**.
48. L. Zhao, L. Härtel, N. Shah, L. Akoglu. *A Practical, Progressively Expressive Graph Neural Network*. **NeurIPS 2022**.
47. Y. Wang, Y. Zhao, N. Shah, T. Derr. *Imbalanced Graph Classification via Graph-of-Graph Neural Networks*. **CIKM 2022**.
46. J. Jiang, N. Murrugarra-Llerena, M. Bos, Y. Liu, N. Shah, L. Neves, F. Barbieri. *Sunshine with a Chance of Smiles: How does Weather Impact Sentiment on Social Media?*. **ICWSM 2022**.
45. S. Zhang, Y. Liu, Y. Sun, N. Shah. *Graph-less Neural Networks: Teaching Old MLPs New Tricks via Distillation*. **ICLR 2022**.
44. W. Jin, L. Zhao, S. Zhang, Y. Liu, J. Tang, N. Shah. *Graph Condensation for Graph Neural Networks*. **ICLR 2022**.
43. L. Zhao, W. Jin, L. Akoglu, N. Shah. *From Stars to Subgraphs: Uplifting Any GNN with Local Structure Awareness*. **ICLR 2022**.
42. Y. Ma, X. Liu, N. Shah, J. Tang. *Is Homophily a Necessity for Graph Neural Networks?*. **ICLR 2022**.
41. W. Jin, X. Liu, X. Zhao, Y. Ma, N. Shah, J. Tang. *Automated Self-Supervised Learning for Graphs*. **ICLR 2022**.
40. X. Tang, Y. Liu, X. He, S. Wang, N. Shah. *Ranking Friend Stories on Social Platforms with Edge-Contextual Local Graph Convolutions*. **WSDM 2022**.
39. S. Sikdar, N. Shah, T. Weninger. *Attributed Graph Modeling with Vertex Replacement Grammars*. **WSDM 2022**.
38. H. Shin, T. Kwon, N. Shah, K. Shin. *Finding a Concise, Precise and Exhaustive Set of Near Bi-Cliques in Dynamic Graphs*. **WSDM 2022**.
37. T. Zhao, B. Ni, W. Yu, Z. Guo, N. Shah, M. Jiang. *Action Sequence Augmentation for Early Graph-based Anomaly Detection*. **CIKM 2021**.
36. Y. Ma, X. Liu, T. Zhao, Y. Liu, J. Tang, N. Shah. *A Unified View on Graph Neural Networks as Graph Signal Denoising*. **CIKM 2021**.
35. E. Gujral, L. Neves, E. Papalexakis, N. Shah. *Niche Detection in User Content Consumption Data*. **CIKM 2021**.
34. S. Shekhar, N. Shah, L. Akoglu. *FairOD: Fairness-aware Outlier Detection*. **AIES 2021**.
33. Q. Yang, W. Wang, L. Pierce, R. Vaish, X. Shi, N. Shah. *Online Communication Shifts in the Midst of the Covid-19 Pandemic: A Case Study on Snapchat*. **ICWSM 2021**.
32. F. A. Chowdhury, Y. Liu, K. Saha, N. Vincent, L. Neves, N. Shah, M. Bos. *CEAM: The Effectiveness of Cyclic and Ephemeral Attention Models of User Behavior on Social Platforms*. **ICWSM 2021**.
31. A. Sankar, Y. Liu, J. Yu, N. Shah. *Graph Neural Networks for Friend Ranking in Large-scale Social Platforms*. **WWW 2021**.
30. K. Saha, Y. Liu, N. Vincent, F. A. Chowdhury, L. Neves, N. Shah, M. Bos. *AdverTiming Matters: Examining User Ad Consumption for Effective Ad Allocations on Social Media*. **CHI 2021**.
29. T. Zhao, Y. Liu, L. Neves, O. Woodford, M. Jiang, N. Shah. *Data Augmentation for Graph Neural Networks*. **AAAI 2021**.
28. B. Joshi, F. Barbieri, N. Shah, L. Neves. *The Devil is in the Details: Evaluating Limitations of Transformer-based Methods for Granular Tasks*. **COLING 2020**.
27. P. Kaghazgaran, M. Bos, L. Neves, N. Shah. *Social Factors in Closed-Network Content Consumption*. **CIKM 2020**.
26. S. Abdali, R. Gurav, S. Menon, D. Fonseca, N. Entezari, N. Shah, E. Papalexakis. *Identifying Misinformation from Website Screenshots*. **ICWSM 2021**.
25. S. Abdali, N. Shah, E. Papalexakis. *Semi-Supervised Multi-aspect Misinformation Detection with Hierarchical Joint Decomposition*. **ECML-PKDD 2020**.
24. X. Tang, Y. Liu, N. Shah, X. Shi, P. Mitra, S. Wang. *Knowing your FATE: Friendship, Action and Temporal Explanations for User Engagement Prediction on Social Apps*. **KDD 2020**.
23. N. Shah. *FARE: Schema-Agnostic Anomaly Detection in Social Event Logs*. **DSAA 2019**.
22. H. Nilforoshan, N. Shah. *SliceNDice: Mining Suspicious Multi-attribute Entity Groups with Multi-view Graphs*. **DSAA 2019**.
21. H. Lamba, N. Shah. *Modeling Dwell Time Engagement on Visual Multimedia*. **KDD 2019**.

20. H. Habib, N. Shah, R. Vaish. *Impact of Contextual Factors on Public Snapchat Sharing*. **CHI 2019. Best Paper Honorable Mention Award.**
19. S. Jain, D. Niranjana, H. Lamba, N. Shah, P. Kumaraguru. *Characterizing and Detecting Livestreaming Chatbots*. **ASONAM 2019.**
18. G. B. Guacho, S. Abdali, N. Shah, E. Papalexakis. *Semi-Supervised Content-based Detection of Misinformation via Tensor Embeddings*. **ASONAM 2018.**
17. N. Gupta, D. Eswaran, N. Shah, L. Akoglu, C. Faloutsos. *Beyond Outlier Detection: LookOut for Pictorial Explanation*. **ECML-PKDD 2018.**
16. N. Shah, H. Lamba, A. Beutel, C. Faloutsos. *The Many Faces of Link Fraud*. **ICDM 2017.**
15. D.-C. Juan, N. Shah, M. Tang, Z. Qian, D. Marculescu, C. Faloutsos. *M3A: Model, MetaModel, and Anomaly Detection in Web Searches*. **DSAA 2017.**
14. N. Shah. *FLOCK: Combating Astroturfing on Livestreaming Platforms*. **WWW 2017.**
13. B. Hooi, H. A. Song, A. Beutel, N. Shah, K. Shin, C. Faloutsos. *FRAUDAR: Bounding Graph Fraud in the Face of Camouflage*. **KDD 2016. Best Paper Award.**
12. B. Hooi, N. Shah, A. Beutel, S. Günnemann, L. Akoglu, M. Kumar, D. Makhija, C. Faloutsos. *BIRDNEST: Bayesian Inference for Ratings-Fraud Detection*. **SDM 2016.**
11. N. Shah, D. Koutra, T. Zou, B. Gallagher, C. Faloutsos. *TimeCrunch: Interpretable Dynamic Graph Summarization*. **KDD 2015.**
10. M. Giatzoglou, D. Chatzakou, N. Shah, A. Beutel, S. Guenneman, C. Faloutsos, A. Vakali. *ND-Sync: Detecting Synchronized Fraud Activities*. **PAKDD 2015.**
9. M. Giatzoglou, D. Chatzakou, N. Shah, C. Faloutsos, A. Vakali. *Retweeting Activity on Twitter: Signs of Fraud*. **PAKDD 2015.**
8. N. Shah, A. Beutel, B. Gallagher, C. Faloutsos. *Spotting Suspicious Link Behavior with fBox: An Adversarial Perspective*. **ICDM 2014.**
7. N. Shah, E. Schendel, S. Pendse, S. Lakshminarasimhan, T. Rogers, N. Samatova. *Improving I/O Throughput with PRIMACY: Preconditioning ID-Mapper for Compressing Incompressibility*. **CLUSTER 2012.**
6. E. Schendel, Y. Jin, N. Shah, J. Chen, C.-S. Chang, S.-H. Ku, S. Ethier, S. Klasky, R. Latham, R. Ross, N. Samatova. *ISOBAR Preconditioner for Effective and High-throughput Lossless Data Compression*. **ICDE 2012.**
5. I. Arkatkar, J. Jenkins, S. Lakshminarasimhan, N. Shah, E. Schendel, S. Ethier, et al. *Analytics-driven Lossless Data Compression for Rapid In-situ Indexing, Storing and Querying*. **DEXA 2012.**
4. Y. Jin, S. Lakshminarasimhan, N. Shah, Z. Gong, C.-S. Chang, J. Chen, et al. *S-preconditioner for Multi-fold Data Reduction with Guaranteed User-controlled Accuracy*. **ICDM 2011.**
3. S. Lakshminarasimhan, N. Shah, S. Ethier, S. Klasky, R. Latham, R. Ross, N. Samatova. *Compressing the Incompressible with ISABELA: In-situ Reduction of Spatio-Temporal Data*. **EuroPar 2011.**
2. N. Shah, Y. Shpanskaya, C.-S. Chang, S.-H. Ku, A. Melechko, N. Samatova. *Automatic and Statistically Robust Spatio-temporal Detection and Tracking of Fusion Plasma Fronts*. **SciDAC 2010.**
1. P. Breimyer, G. Kora, W. Hendrix, N. Shah, N. Samatova. *pR: Automatic Parallelization of Data-parallel Statistical Computing Codes for R in Hybrid Multi-node and Multi-core Environments*. **IADIS 2009.**

Journal Publications

11. Z. Guo, T. Zhao, Y. Liu, K. Dong, W. Shiao, N. Shah, N. Chawla. *Node Duplication Improves Cold-start Link Prediction*. **TMLR 2025.**
10. Y. Dong, W. Shiao, Y. Liu, J. Li, N. Shah, T. Zhao. *Do Graph Neural Networks Improve Node Representation Learning for All?*. **DMLR 2025.**
9. F. Xia, R. Lambiotte, N. Shah, H. Tong, I. King. *Guest Editorial: Special Issue on Graph Learning*. **IEEE TNNLS 2024.**
8. D. Gomez-Zara, Y. Liu, L. Neves, N. Shah, M. Bos. *Unpacking the Exploration-Exploitation Tradeoff on Snapchat*. **Computers in Human Behavior 2023.**
7. T. Zhao, T. Jiang, N. Shah, M. Jiang. *A Synergistic Approach for Graph Anomaly Detection with Pattern Mining and Feature Learning*. **IEEE TNNLS 2021.**
6. Y. Liu, T. Safavi, N. Shah, D. Koutra. *Reducing Large Graphs to Small Supergraphs: A Unified Approach*. **SNAM**

2018.

5. B. Hooi, K. Shin, H. A. Song, A. Beutel, N. Shah, C. Faloutsos. *Graph-based Fraud Detection in the Face of Camouflage*. **TKDD 2017**.
4. N. Shah, D. Koutra, L. Jin, T. Zou, B. Gallagher, C. Faloutsos. *On Summarizing Large-Scale Dynamic Graphs*. **Data Engineering Bulletin 2017**.
3. D. Koutra, N. Shah, J. T. Vogelstein, B. Gallagher, C. Faloutsos. *DeltaCon: A Principled Massive-Graph Similarity Function with Attribution*. **TKDD 2015**.
2. J. Jenkins, I. Arkatkar, S. Lakshminarasimhan, D. Boyuka, E. Schendel, N. Shah, et al. *ALACRITY: Analytics-driven Lossless Data Compression for Rapid In-situ Indexing, Storing, and Querying*. **TLDKS 2013**.
1. S. Lakshminarasimhan, N. Shah, S. Ethier, S. Klasky, R. Latham, R. Ross, N. Samatova. *ISABELA for Effective In-situ Compression of Scientific Data*. **Concurrency and Computation: Practice and Experience 2011**.

Workshop Publications

9. *Improving Out-of-Vocabulary Handling in Recommendation Systems*. **TheWebConf RelWeb 2025**.
8. M. Kolodner, M. Ju, Z. Fan, T. Zhao, E. Ghazizadeh, Y. Wu, N. Shah, Y. Liu. *Robust Training Objectives Improve Embedding-based Retrieval in Industrial Recommendation Systems*. **RecSys RobustRecSys 2024**.
7. M. Ju, T. Zhao, W. Yu, N. Shah, Y. Ye. *GraphPatcher: Mitigating Degree Bias for Graph Neural Networks via Test-time Augmentation*. **TheWebConf DCAI 2024**.
6. P. Kung, Z. Fan, T. Zhao, Y. Liu, L. Lai, J. Shi, Y. Wu, N. Shah, J. Yu. *Improving Embedding-Based Retrieval in Friend Recommendation with ANN Query Expansion*. **TheWebConf DCAI 2024**.
5. N. Shah. *Scale-Free, Attributed and Class-Assortative Graph Generation to Facilitate Introspection of Graph Neural Networks*. **KDD MLG 2020**.
4. R. Kumar, M. Kumar, N. Shah, C. Faloutsos. *Did We Get It Right? Predicting Query Performance in E-commerce Search*. **SIGIR eCom 2018**.
3. Y. Liu, T. Safavi, N. Shah, D. Koutra. *Reducing Million-Node Graphs to a Few Structural Patterns: A Unified Approach*. **KDD MLG 2016**.
2. N. Shah, A. Beutel, B. Hooi, L. Akoglu, S. Günnemann, D. Makhija, M. Kumar, C. Faloutsos. *EdgeCentric: Anomaly Detection in Edge-Attributed Networks*. **ICDM DMCS 2016**.
1. Y. Liu, N. Shah, D. Koutra. *An Empirical Comparison of the Summarization Power of Graph Clustering Methods*. **NIPS NSIS 2015**.

Surveys

5. S. Kim, G. Lee, K. Kim, L. Collins, N. Shah, K. Shin. *Retrieval-Augmented Generation for LLM-based Recommender Systems: A Comprehensive Survey*. **KDD 2026**.
4. S. Kim, G. Lee, K. Kim, L. Collins, N. Shah, K. Shin. *Retrieval-Augmented Generation for LLM-based Recommender Systems: A Comprehensive Survey*. **CIKM 2026**.
3. H. Han, Y. Wang, H. Shomer, K. Guo, J. Ding, Y. Lei, M. Halappanavar, R. A. Rossi, et al. *Retrieval-Augmented Generation with Graphs (GraphRAG)*. **arXiv 2024**.
2. T. Zhao, W. Jin, Y. Liu, Y. Wang, G. Liu, S. Günnemann, N. Shah, M. Jiang. *Graph Data Augmentation for Graph Machine Learning*. **IEEE Data Engineering Bulletin 2023**.
1. S. Kumar, N. Shah. *False Information on the Web and Social Media*. **arXiv 2018**.

Tutorials

5. Y. Liu, T. Zhao, M. Kolodner, K. Montemayor, S. Vij, N. Shah. *Training Industry-Scale Graph Neural Networks with GiGL*. **KDD 2025**.
4. R. Xue, H. Han, T. Zhao, N. Shah, J. Tang, X. Liu. *Large-Scale Graph Neural Networks: The Past and New Frontiers*. **AAAI 2024**.
3. R. Xue, H. Han, T. Zhao, N. Shah, J. Tang, X. Liu. *Large-Scale Graph Neural Networks: The Past and New Frontiers*. **SDM 2024**.

2. R. Xue, H. Han, T. Zhao, N. Shah, J. Tang, X. Liu. *Large-Scale Graph Neural Networks: The Past and New Frontiers*. **KDD 2023**.
1. T. Zhao, K. Ding, W. Jin, G. Liu, M. Jiang, N. Shah. *Augmentation Methods for Graph Learning*. **SDM 2023**.

Book Chapters

3. S. Abdali, G. Bastidas, N. Shah, E. Papalexakis. *Tensor Embeddings for Content-Based Misinformation Detection with Limited Supervision*. In: *Disinformation, Misinformation, and Fake News in Social Media*.
2. N. Shah. *Introduction to R*. In: *Practical Graph Mining with R*.
1. K. Padmanabhan, S. Lakshminarasimhan, Z. Gong, J. Jenkins, N. Shah, E. Schendel, I. Arkatkar, R. Ross, S. Klasky, N. Samatova. *In-situ Analysis in Support of Exploratory Scientific Discovery in Data-Intensive Science*. In: *Data-Intensive Science*.

ACADEMIC EXPERIENCE

Invited Talks

- Panelist, *AI-Augmented Academia: Research Innovations and Career Planning for Tomorrow's Workforce*, KDD Doctoral Consortium, 2025
- Panelist, *From Lists to Dialogues: Rethinking Personalization with Generative AI*, KDD Workshop on Generative AI for Personalization, 2025
- Panelist, *From Research to Product*, SIGIR Workshop on E-Commerce, 2025
- Panelist, *Is Search Dead? The Rise or Demise of Search in the Era of LLMs*, SIGIR, 2025
- Panelist, UCR exploreCSR Workshop, 2024
- Keynote Speaker, KDD Undergraduate Consortium, 2024
- Panelist, KDD Graph Learning Benchmarks (GLB) Workshop, 2023
- Invited Speaker, Samsung Research, 2023
- Keynote Speaker, RE-WORK AI Summit West, 2023
- Panel Moderator, KDD Misinformation and Misbehavior (MIS²-TrueFact) Workshop, 2022
- Panelist, KDD Deep Learning on Graphs (DLG) Workshop, 2022
- Keynote Speaker, KDD Deep Learning on Graphs (DLG) Workshop, 2022
- Panel Moderator, TigerGraph AI Summit, 2022
- Invited Speaker, WSDM, 2022
- Panelist, The Knowledge Graph Conference, 2022
- Invited Speaker, The Knowledge Graph Conference, 2022
- Invited Speaker, UC Riverside CSE Colloquium, 2022
- Keynote Speaker, WSDM Machine Learning on Graphs (MLoG) Workshop, 2022
- Panelist, UCR exploreCSR Workshop, 2022
- Invited Speaker, Pinterest Trust and Safety Summit, 2021
- Panel Moderator, KDD Outlier Detection and Discovery (ODD) Workshop, 2021
- Keynote Speaker, KDD Machine Learning in Finance (MLF) Workshop, 2021
- Keynote Speaker, SDM Doctoral Consortium, 2021
- Keynote Speaker, SDM Minisymposium on Dynamic Networks, 2020
- Keynote Speaker, ICDM Doctoral Consortium, 2019
- Keynote Speaker, WWW CyberSafety Workshop, 2018
- Keynote Speaker, KDD Outlier Detection De-constructed Workshop, 2018
- Keynote Speaker, ECML-PKDD PhD Forum, 2018

Service

Conference Organization

- Program Chair, CODS-COMAD, 2025

- Sponsorship Chair, SDM, 2025
- Hands-on Tutorial Chair, KDD, 2023, 2024
- Sponsorship Chair, ICWSM, 2023
- Cup Chair, WSDM, 2020, 2022
- Program Chair, ASONAM, 2019
- Organizer, KDD Generative AI for Recommender Systems and Personalization (GenAI4RecP) Workshop, 2025
- Organizer, KDD Federated Learning with Graph Data (FedGraph) Workshop, 2024, 2025
- Organizer, ICDM Mining and Learning on Graphs (MLoG) Workshop, 2022, 2023
- Organizer, KDD Mining and Learning with Graphs (MLG) Workshop, 2022, 2023
- Organizer, CIKM Federated Learning with Graph Data (FedGraph) Workshop, 2022
- Organizer, KDD Misinformation and Misbehavior Mining (MIS²) Workshop, 2021, 2022
- Organizer, TheWebConf CyberSafety Workshop, 2019, 2020
- Session Chair, TheWebConf (Security and Privacy, 2018; Graph Models, 2021)
- Session Chair, KDD (Graph Algorithms, 2020; Graphs and Networks, 2021; Graph Learning and Social Network, 2022)
- Session Chair, DSAA (Subgraphs, 2019)
- Session Chair, ICDM (Social track, 2016)

Peer Review

- Area Chair, NeurIPS, 2025
- Area Chair, ICLR, 2025, 2026
- Area Chair, KDD, 2022, 2025, 2026
- Area Chair, TheWebConf, 2024, 2025, 2026
- Area Chair, LoG, 2022, 2023, 2024, 2025
- Senior Program Committee, AAAI, 2023, 2024, 2025, 2026
- Senior Program Committee, PAKDD, 2023
- Senior Program Committee, SDM, 2022–2025
- Senior Program Committee, WSDM, 2022–2025
- Senior Program Committee, CIKM, 2021–2025
- Early Career Data Mining Award Committee, SDM, 2023, 2024
- Best Paper Award Committee, SDM, 2023, 2024
- Program Committee, ICLR, 2024
- Program Committee, NeurIPS, 2023, 2024
- Program Committee, ICDM, 2022
- Program Committee, ASONAM, 2022
- Program Committee, WSDM, 2019–2021
- Program Committee, KDD, 2019–2021
- Program Committee, TheWebConf, 2015, 2018, 2020–2022
- Program Committee, SDM, 2018–2021
- Program Committee, CIKM, 2017, 2020
- Reviewer, ACM TKDD, 2018–2020; Springer DAMI, 2018, 2019; ACM TSOC, 2018, 2019; ACM TKDE, 2016, 2017
- Reviewer, CSCW, 2019; CHI, 2019
- Reviewer, WISE, 2014; IPDPS, 2011
- Program Committee, numerous workshops (MLG, GLB, HeteroNAM, MIS², GreS, PhD Forums, Demo Sessions) at KDD, WSDM, WWW, ICML, ICDM, RecSys, CIKM

Mentorship

Internships and Student Advisory Roles

- Jingzhe Liu (intern at Snap Research, 2025)
- Geon Lee (intern at Snap Research, 2025)
- Kulin Shah (intern at Snap Research, 2025)

- Xingyue Huang (intern at Snap Research, 2025)
- Xueying Ding (intern at Snap Research, 2025)
- Ngoc Bui (intern at Snap Research, 2024)
- Xinyi Wu (intern at Snap Research, 2024)
- Donald Loveland (intern at Snap Research, 2024)
- Runjin Chen (intern at Snap Research, 2024)
- Jing Zhu (intern at Snap Research, 2024)
- Haitao Mao (intern at Snap Research, 2024)
- Agostina Calabrese (intern at Snap Research, 2023)
- Yushun Dong (intern at Snap Research, 2023)
- Vijay Prakash Dwivedi (intern at Snap Research, 2023)
- Mingxuan Ju (intern at Snap Research, 2023)
- Zhichun Guo (intern at Snap Research, 2022)
- Vedant Bhatia (intern at Snap, 2022)
- Yiwei Wang (intern at Snap Research, 2022)
- William Shiao (intern at Snap Research, 2022 & 2023)
- Xiaotian Han (intern at Snap Research, 2022)
- Cazamere Comrie (intern at Snap Research, 2021)
- Lingxiao Zhao (intern at Snap Research, 2021)
- Wei Jin (intern at Snap Research, 2021 & 2022)
- Shichang Zhang (intern at Snap Research, 2021)
- Yingtong Dou (intern at Snap Research, 2021)
- Yozen Liu (RA at Snap Research, 2020)
- Qi Yang (intern at Snap Research, 2020)
- Satyaki Sikdar (intern at Snap Research, 2020)
- Yao Ma (intern at Snap Research, 2020)
- Aravind Sankar (intern at Snap Research, 2020)
- Tong Zhao (intern at Snap Research, 2020)
- Nicholas Vincent (intern at Snap Research, 2020)
- Farhan Asif Chowdhury (intern at Snap Research, 2020)
- Koustuv Saha (intern at Snap Research, 2020)
- Brihi Joshi (intern at Snap Research, 2019)
- Shiyang Yan (intern at Snap Research, 2019)
- Xianfeng Tang (intern at Snap Research, 2019 & 2020)
- Parisa Kaghazgaran (intern at Snap Research, 2019)
- Himel Dev (intern at Snap Research, 2019)
- Anis Zaman (intern at Snap Research, 2019)
- Can Liu (intern at Snap Research, 2019)
- Dipankar Niranjana (BS student, IIIT Delhi, 2018)
- Shreya Jain (BS student, IIIT Delhi, 2018)
- Hamed Nilforoshan (intern at Snap Research, 2018)
- Hana Habib (intern at Snap Research, 2018)
- Hemank Lamba (intern at Snap Research, 2018)
- Rohan Kumar (visiting CS student at CMU, 2017)
- Qicheng Huang (EE PhD student at CMU, 2017)
- Chenlei Fang (EE PhD student at CMU, 2017)
- Tianmin Zou (CS MS student at CMU, 2017)

Thesis Supervision

- Committee Member, Haitao Mao, 2024
- Committee Member, Harry Shomer, 2024
- Committee Member, Lingxiao Zhao, 2024
- Committee Member, William Shiao, 2024

- Committee Member, Zhichun Guo, 2023
- Committee Member, Wei Jin, 2023
- Committee Member, Yingtong Dou, 2021
- Committee Member, Aravind Sankar, 2021
- Committee Member, Tong Zhao, 2020

Funding

- Contributed towards Flipkart faculty grant (with CMU: Bryan Hooi, Dhivya Eswaran, Christos Faloutsos)
- Contributed towards Wharton Customer Analytics Initiative proposal *Fraud Detection through Mining Dynamic Behavior for Group Anomalies* (with CMU: Alex Beutel, Christos Faloutsos)
- Contributed towards PNC Center for Financial Services proposal PF15003: *Fraud Detection in Financial Data* (with CMU: Alex Beutel, Christos Faloutsos)
- Contributed towards DOE-NNSA-30788.1.1990222: *Quantifying Network Changes* (with CMU: Danai Koutra, Christos Faloutsos)
- Contributed towards NSF IIS-1028746: *Collaborative Research: Understanding Climate Change — A Data Driven Approach* (with NCSU: Nagiza Samatova, Fredrick Semazzi)

Teaching

- Guest Lecture, *Improving the Scalability of Graph Neural Networks*, CMU 11-741 Machine Learning for Text and Graph-based Mining (Prof. Yiming Yang), 2024
- Seminar Talk, *Improving the Scalability of Graph Neural Networks*, Brandeis University Machine Learning Seminar, 2024
- Guest Lecture, *Improving the Scalability of Graph Neural Networks*, Georgia Tech ISYE 4803 Network Science and Analysis (Prof. Tejas Santanam), 2023
- Guest Lecture, *Improving the Scalability of Graph Neural Networks*, Georgia Tech CSE 6240 Web Search and Text Mining (Prof. Srijan Kumar), 2023
- Guest Lecture, *Improving the Scalability of Graph Neural Networks*, Michigan State CSE 482 Big Data Analysis (Prof. Jiliang Tang), 2022
- Guest Lecture, *Machine Learning on Graphs with Scarce Labels*, RPI MGMT-6560-02 Introduction to Machine Learning Applications (Prof. Lydia Manikonda), 2021
- Guest Lecture, *Mining Misbehavior on Large-Scale Social Platforms*, Vanderbilt CS-5981-06 Social Network Analysis (Prof. Tyler Derr), 2020
- Guest Lecture, *A Foray into Graph Mining*, USC CSCI-699 Introduction to Information Extraction (Prof. Xiang Ren), 2019
- Guest Lecture, *Graph Mining for Fraud Detection*, CMU 15-300 Research and Innovation in Computer Science (Prof. Todd Mowry), 2015
- Teaching Assistant, CMU 15-300 Research and Innovation in Computer Science (Prof. Todd Mowry), 2015
- Teaching Assistant, CMU 15-826 Multimedia Databases and Data Mining (Prof. Christos Faloutsos), 2014

TECHNICAL SKILLS

- *Cloud Platforms*: GCP, AWS
- *Languages/Tools*: Python, Java, C, C++, x86 Assembly
- *Web*: SQL, HTML, PHP, JavaScript, CSS, WordPress
- *Engineering Tools*: MATLAB, R
- *Typesetting*: LaTeX, Microsoft Office, LibreOffice
- *Version Control*: Git, Subversion
- *Operating Systems*: macOS, Ubuntu Linux, Microsoft Windows

REFERENCES

Available upon request.